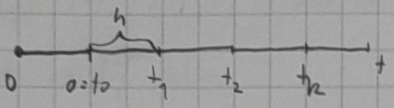
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Dors 9

Euler yontemi

$y' = f(t,y) \qquad a \le t \le b \qquad y(a) = \alpha$



N parcaya bol $\qquad b = t_N$

$h = \frac{b-a}{2} \qquad t_k = a + kh \qquad k = 0, 1, 2 \dots N$

$t_k, \; k = 0, 1, \dots N-1$

$y \in C^2[a,b] \rightarrow y(t) = y(t_k) + y'(t_k)\cdot(t - t_k) + \frac{y''(\xi)}{2}(t - t_k)^2$

$t = t_{k+1}$

$y(t_{k+1}) = y(t_k) + y'(t_k)\,\underbrace{(t_{k+1} - t_k)}_{h} + \frac{y''(\xi_k)}{2}\,\underbrace{(t_{k+1} - t_k)}_{h}^2$

$w_0 = \alpha$

$w_{k+1} = w_k + h f(t_k, w_k)$

$y(t_{k+1}) = y(t_k) + y'(t_k)\cdot h + \frac{y''(\xi_k)\cdot h^2}{2}$

$y(t_0) = \alpha$

6) $y' = 1 + \frac{y}{t} \qquad 1 \le t \le 2 \qquad y(1) = 2 \qquad$ 5 dp cozum $\qquad h = 0{,}25$ olsun

$f(t,y) = 1 + \frac{y}{t} \qquad a = 1 \quad b = 2 \quad \alpha = 2 \qquad h = \frac{b-a}{N} \qquad N = 4$

$w_0 = 2$

$w_1 = w_0 + 0{,}25 \cdot \left(1 + \frac{w_0}{t_0}\right) = 2{,}75$

$w_2 = w_1 + 0{,}25\left(1 + \frac{w_1}{t_1}\right) =$

$w_3 = w_2 + 0{,}25\left(1 + \frac{w_2}{t_2}\right) =$

$w_4 = w_3 + 0{,}25\left(1 + \frac{w_3}{t_3}\right) =$

$y(t) = 2x + x\ln(x)$

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Yüksek dereceden Taylor yöntemi

$y' = f(t,y) \quad a \le t \le b$

$y(a) = \alpha \qquad y(t_k) = w_k \qquad w_0 = \alpha \qquad w_{k+1} = w_k + h f(t_k, w_k) \qquad k = 0 \dots N-1$

Local hata, $y(t_k) = y_k \qquad k = 0, \dots N-1$

$$\tau_{k+1}(h) = \frac{y_{k+1} - y_k}{h} - y'(t_k) = \frac{y''(\xi)}{2!} \cdot h \qquad w_{k+1} = w_k + h T^n(t_k, w_k)$$

$$y(t_{k+1}) = y(t_k) + h y'(t_k) + \frac{h^2}{2!} y'(t_k) + \dots + \frac{h^n}{n!} y^n(t_k)$$

Ö/ $y' = y - t^2 + 1 \qquad 0 \le t \le 1 \qquad y(0) = 0.5$ BDP 2. dereceden Taylor formülü

adım boyu $h = 0,25$ olarak veriliyor.

$f(t,y) = y - t^2 + 1 \qquad a = 0 \quad b = 1 \quad \alpha = 0,5 \quad h = 0,25 \quad \frac{d}{dt} f(t, y(t)) = y' - 2t$

$= y - t^2 - 2t + 1$

$N = 4$

$t_k = 0,25k \qquad k = 0,1,2,3$

$T^2(t_k, w_k) = f(t_k, w_k) + \frac{h}{2} \frac{d}{dt} f(t_k, w_k)$

$= w - t_k^2 + 1 + \frac{h}{2}(w_k - t_k^2 - 2t_k + 1)$

$k = 0 \quad w_1 = 0,9219$

$k = 1 \quad w_2 = 1,4292$

$k = 2 \quad w_3 = 2,0109$

$k = 3 \quad w_4 = 2,6526$

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Pos 10

Matrislerin Çarpanlarına Ayrılması

1) L U faktorizasyonu

2) Cholesky (veya LL^T) faktorizasyonu

$n \times n$, tekil olmayan ($\det A \neq 0$) $A[a_{ij}]$ matrisi verilmiş olsun. Her n boyutlu sütun b vektörü için homojen olmayan matris denklemi $Ax = b$ veya $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Gauss Eleminasyonu $\det \neq 0$ gerektirmez

Gauss kuralı $\det \neq 0$ gerektirir.

LU faktorizasyonu

$A = [a_{ij}]$ matrisini $L = [l_{ij}]$ alt üçgen, $U = [u_{ij}]$ üst üçgen $A = LU$ şeklinde çarpanlara ayrılmasına LU faktorizasyonu denir. Genelliği bozmadan L matrisi köşegeni 1 olacak şekilde kabul edebiliriz. LU bilinsin varsayalım;

$Ax = b \quad LUx = b \quad Ux = y \quad Ly = b$

$Ly = b \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ l_{21} & 1 & 0 & 0 \\ l_{31} & l_{32} & 1 & 0 \\ l_{41} & l_{42} & l_{43} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$ $y_1 = b_1$, $y_2 = b_2 - l_{21} y_1$ } Forward substitution

$Ux = y \Rightarrow \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$ } Backward sub.

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$A = LU = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

$u_{11} = a_{11} \quad u_{12} = a_{12} \quad u_{13} = a_{13}$

$a_{21} = u_{11} \cdot l_{21}$

Verilmiş A matrisinin a) LU faktoru bul

b) Bunu kullanarak $A_x = [2, 3, -1]^T$ sistemini çözün (T → transpoz)

a)

$A = \begin{bmatrix} 3 & 1 & 2 \\ 6 & 3 & 3 \\ -3 & 2 & -3 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -2 \end{bmatrix}}_{U}$

b)

$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}}_{L} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$

$y_1 = 2$

$y_2 = -1$

$y_3 = 4$

$\underbrace{\begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}}_{U} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}}_{y_1, y_2, y_3}$

$x_1 = -1$

$x_2 = 1$

$x_3 = 2$

2) Cholesky faktorizasyonu: A Matrisi simetrik ve Pozitif tanımlı olmalı. (Sadece bu)

$x^T Ax > 0$ olmalı $(x_1 x_2 x_3)$ $\quad A = A^T$ $\quad a_{ij} = a_{ji}$

a)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad x^T Ax = [x_1\ x_2\ x_3] \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2 > 0$$

gibi

Daha büyük matrislerde pozitiflik testi

$n \times n$ boyutlu $A = [a_{ij}]$ matrisi için, $A_k = \begin{bmatrix} a_{11} & \cdots \\ \vdots & \ddots \\ & \end{bmatrix}$ $1 \le k \le n$ — baş köşegen matrisi

A matrisinin pozitif tanımlı olması için baş köşegen matrislerinin $\det A_k > 0$ olmalı gerek ve yeterdir.

Cholesky Faktorizasyon

$$A = L_1 L^T \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & & \end{bmatrix}$$

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Ø $A = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 10 & 3 \\ 2 & 3 & 5 \end{bmatrix} = A^T$

a) Apozitív/m: $\det A_1 > 0,\ \det A_2 > 0,\ \det A_3 > 0$

b) cholesky fak. tétszerem ugyanúgy

c) $A_x = [4,5,14]^T$ tanmenti zöm

$$A = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{21}l_{31} + l_{22}l_{32} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}}_{L^T}$$

$$A_x = L\underbrace{L^T x}_{y} = Ly = \begin{bmatrix} 4 \\ 5 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$y_1 = 2$
$y_2 = -1$
$y_3 = 6$

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$$

$x_1 = 1$
$x_2 = -1$
$x_3 = 3$